Supplemental Material for "Lattice refinement followed by the rodeo algorithm: a gate-level benchmark of eigenstate-preparation routes on the Hubbard chain"

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This Supplemental Material collects the verification material behind the main text: the sufficient adiabatic time and the response norm of the path (Sec. S1), the dimer-seeded hierarchy and the composition of steps (Sec. S2), the plaquette spectra (Sec. S3), the variational tables (Sec. S4), the Trotterized refinement at eight sites (Sec. S5), the exact-ensemble filter (Sec. S6), the ingredients and validation of the thirty-two-qubit study (Sec. S7), and the operating-point tables (Sec. S8). Equation, section, table and figure numbers without the prefix S refer to the main text; all numbers are outputs of the notebook `notebooks/ResolutionRefHubbard.ipynb`.

S1 The sufficient adiabatic time and the response norm of the path

Table S1 evaluates the sufficient adiabatic time of Jansen, Ruskai and Seiler [1] in the form used in paper I, $T_b = \epsilon^{-1/2}\int_0^1[\|\ddot{H}\|/\Delta^2 + 7\|\dot{H}\|^2/\Delta^3]\,ds$ with $\|B-A\|$ the spectral norm of $H_{\rm high} - PH_{\rm low}P^\dagger$ in the sector, once with the sector gap and once with the gap in the symmetry sector of the tracked state (Fig. 3), against the empirical T^*. The symmetry-resolved bound is smaller by a factor 5 to 17 but remains three to five orders of magnitude above $T^* = 10/t$, as the pairing bound did (there 3×10^2); the larger ratio here reflects $\|B-A\| = 11$–20, set by the shift μ and the width of the Hubbard spectrum. The bound guarantees a worst-case operator-norm error for every initial state in the branch, T^* records one overlap for one initial state, so the ratio measures the slack of the criterion, not an error in either.

The first-order response norm of the path, $\chi_u(s) = \sum_{n>0}|\langle n(s)|\partial_u H|0(s)\rangle|^2/[E_n(s)-E_0(s)]^2$ with $u=\sin^2(\pi s/2)$ and $\partial_u H = (H_{\rm high}-\mu) - P(H_{\rm low}-\mu)P^\dagger$, is the quantity whose integral along the path sets the adiabatic leakage at leading order; for the pairing model paper I obtained it in closed form at the start of the path, $\chi_u(0) = g^2/\mu_{\rm buf}^2$. Figure S1 (left) follows it along Path A for the chain $4\to8$: $\chi_u(0) = 0.375, 0.527, 0.715$ and 0.874 for $U/t = 0, 2, 4, 8$, the maximum sits at the initial endpoint for every coupling, and the norm falls by a factor two to four along the path (at $U = 8t$ it rises again toward $s=1$, to 0.45, as $\Delta_{\rm eff}$ shrinks). The shift fixes the initial value as it did for the pairing model, and the growth of $\chi_u(0)$ with U is modest, in line with the flat T^*. The right panel compares, with exact slices, the ordinary adiabatic ramp from the free-fermion determinant with the two refinement paths at $U = 4t$: the ramp reaches $p = 0.96878$, 0.99686, 0.99960 and 0.99989 at $T = 4, 8, 12, 16$ (and 0.99999 at $24/t$).

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Table S1: Sufficient adiabatic time T_b (infidelity $\epsilon = 10^{-2}$) along Path A ($\mu_{\mathrm{buf}} = 2t$) for the chain $4 \to 8$, evaluated with the sector gap and with the gap in the symmetry sector of the tracked state, against the empirical T^* ($1 - p < 10^{-3}$). Times in units of $1/t$.

U/t	$\min \Delta$	$\min \Delta_{\mathrm{eff}}$	$\|B - A\|$	$T_b(\Delta)$	$T_b(\Delta_{\mathrm{eff}})$	T^*
0	0.532	0.879	11.1	5.7×10^4	1.2×10^4	10
2	0.319	0.780	11.8	2.5×10^5	2.0×10^4	10
4	0.223	0.588	13.9	1.0×10^6	6.4×10^4	10
8	0.137	0.366	20.0	1.0×10^7	5.8×10^5	10

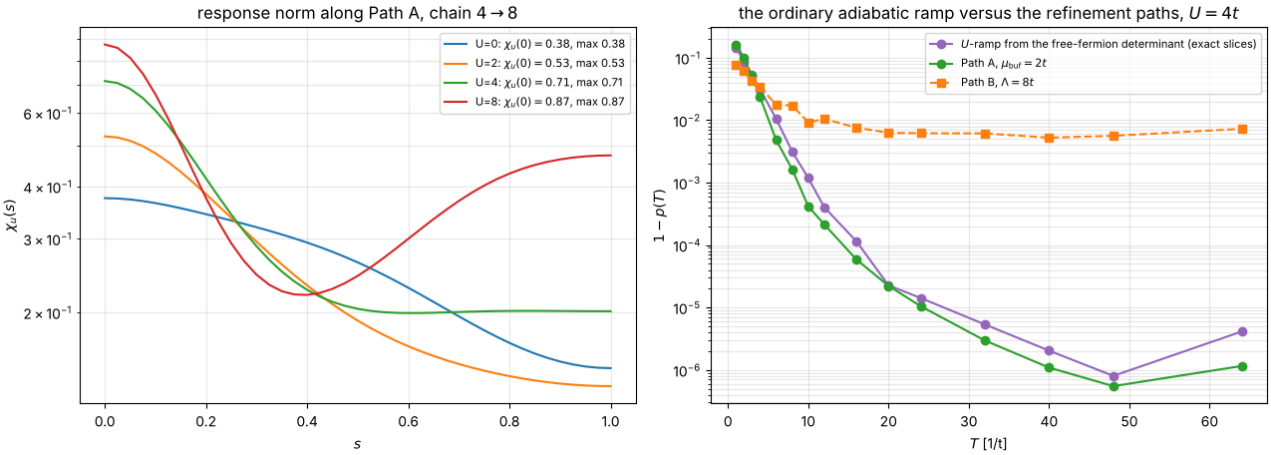


Figure S1: Left: the first-order response norm $\chi_u(s)$ along Path A ($\mu_{\mathrm{buf}} = 2t$) for the chain $4 \to 8$ at $U/t = 0, 2, 4, 8$; the maximum sits at the initial endpoint for every coupling. Right: $1-p(T)$ with exact slices for the ordinary adiabatic ramp from the free-fermion determinant, for Path A and for Path B ($\Lambda = 8t$) at $U = 4t$.

S2 The dimer-seeded hierarchy and the composition of steps

Table S2 shows the first step of the one-dimensional hierarchy, from the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state, $E_{\mathrm{low}} = U - 4t$) to the four-site chain: the doubly-occupied seed is a worse start as U grows, $p_0 = 0.80$ at $U = 0$ down to 0.48 at $U = 8t$, but the effective gap is large ($\Delta_{\mathrm{eff}} = 2.2t$–$0.8t$) and $p > 0.9997$ is reached by $T = 16/t$ at every U. Figure S2 composes the steps (Sec. 4.3): with the refined four-site state as the input of the $4 \to 8$ step and the shift referenced to its variational energy $\langle\psi|H_{\mathrm{low}}|\psi\rangle$, the chain $2 \to 4 \to 8$ at $U = 4t$ with $T = 16/t$ per step reaches $p = 0.99988$ against 0.99994 from the exact coarse input; with $T = 8/t$ in the first step the second saturates at $p = 0.9979$, the overlap of its input. The single-shot refinement $2 \to 8$ with blocks of four fine sites starts from $p_0 = 0.579$ and reaches $p = 0.9966$ at $T = 16/t$ and 0.99988 only at $T = 32/t$.

The step $8 \to 16$ at $(2, 2)$, $U = 4t$ ($784 \to 14\,400$ states) starts from $p_0 = 0.869$ with $E_{\mathrm{low}} = -6.817213$ and $E_{\mathrm{high}} = -7.209813$. Among the forty lowest eigenstates of the fine sector the first to carry weight lies at $\Delta_{\mathrm{eff}} = 0.138t$ (weight 10^{-4}), the next at $0.415t$ and $0.505t$ (the latter with weight 10^{-2}), while the eight other states below $0.4t$, including the one that defines the sector gap ($0.051t$), carry none. With $\mu_{\mathrm{buf}} = 2t$ the overlap reaches 0.9924, 0.99931 and 0.99995 at $T = 8, 16, 32$; composing three steps from the filled dimer, $2 \to 4 \to 8 \to 16$ with $T = 16/t$ each and the refined state as the input of every step, gives $p = 0.99923$.

S3 The plaquette

Table S3 collects the plaquette refinements of Sec. 4.4. At $(2, 2)$ the fine ground state has weight 0 on the prolonged coarse ground state (B_1 singlet), 0.647 on the prolonged A_2 triplet, and 0.007 on all other coarse states together ($U = 4t$); the lowest state of the $(A_2, S = 1)$ sector that carries weight lies $1.77t$ above the ground state (among the thirty lowest eigenstates). The $U = 0$ point is degenerate in both models and its labels are not meaningful.

Table S2: The first step of the one-dimensional hierarchy: the filled dimer ($N_\uparrow = N_\downarrow = 2$, one state) refined to the four-site open chain (36 states), Path A with $\mu_{\mathrm{buf}} = 2t$.

U/t	E_{low}	E_{high}	p_0	Δ	Δ_{eff}	$p(8)$	$p(16)$
0	-4	-4.4721	0.805	1.236	2.236	0.9999	1.0000
2	-2	-2.8759	0.754	0.789	1.851	0.9995	1.0000
4	0	-1.9531	0.645	0.540	1.358	0.9979	0.9999
8	4	-1.1172	0.480	0.311	0.805	0.9787	0.9997

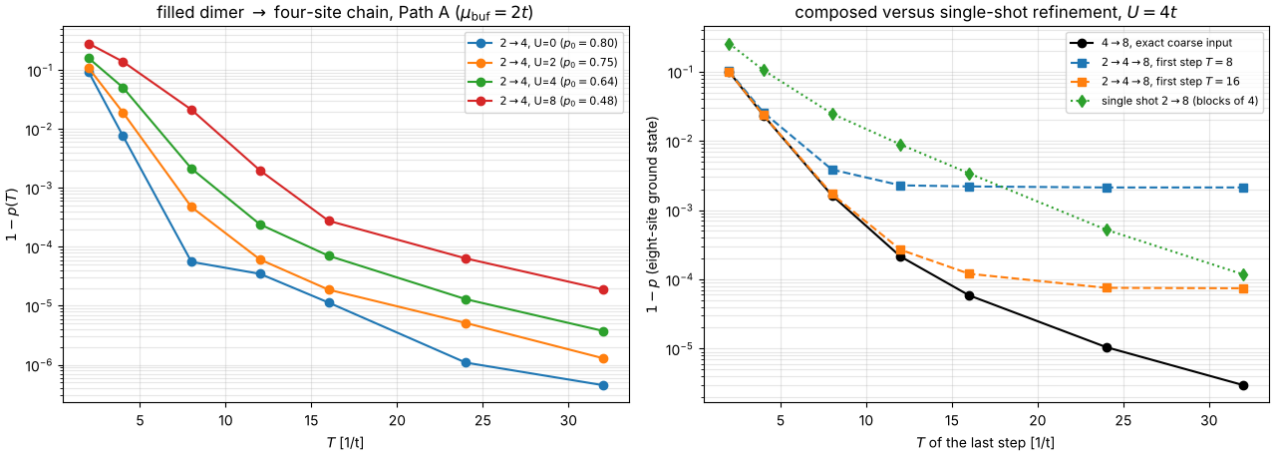


Figure S2: Left: $1-p(T)$ for the dimer step $2 \to 4$ at $U/t = 0, 2, 4, 8$. Right, $U = 4t$: the step $4 \to 8$ from the exact coarse ground state (circles), the composed chain $2 \to 4 \to 8$ with the refined four-site state as input after a first step of $T = 8/t$ or $16/t$ (squares), and the single-shot refinement $2 \to 8$ with blocks of four fine sites (diamonds), all versus the time of the last step.

S4 The variational inputs

Table S4 collects the single-layer UCCSD results. On the coarse chain the 26-parameter ansatz is exact at $U = 0$ and reaches infidelities 1.8×10^{-4}, 3.3×10^{-3} and 1.7×10^{-2} at $U/t = 2, 4, 8$ from a free-fermion reference that is very poor at strong coupling ($E_{\mathrm{HF}} - E_{\mathrm{FCI}} = 2.3t$ at $U = 8t$); on the fine chain the 198-parameter ansatz does slightly better, 4.7×10^{-4}, 3.1×10^{-3}, 1.2×10^{-2}, at 3.1×10^4 CNOTs against 1.7×10^3. Used as the coarse solver, the UCCSD state is prolonged with essentially the same p_0 as the exact coarse state but the refinement then saturates at $p = 0.99669$ ($U = 4t$) and 0.98234 ($U = 8t$) instead of 0.9999. At $U = 4t$, 72 of the 198 fine-chain amplitudes exceed 10^{-2} and 98 exceed 10^{-3}; keeping those 98 (12 singles, 86 doubles) and re-optimizing from the converged values gives $E = -5.618755$ and $1 - F = 3.145 \times 10^{-3}$ against -5.618758 and 3.143×10^{-3} for the full pool, at 15 120 against 30 864 CNOTs. At sixteen sites the single-layer ansatz has 1 022 parameters ($E = -7.197325$ against $E_0 = -7.209813$, $1 - F = 6.98 \times 10^{-3}$, 335 360 CNOTs) and its pruned version 255 parameters (7.00×10^{-3}, 87 776 CNOTs).

ADAPT-VQE (Sec. 3.5) is run with gradient tolerance 10^{-6} and at most 200 operators, every amplitude re-optimized by L-BFGS warm-started from the previous iteration, with the exact ground state used only to record the infidelity. At eight sites ($U = 4t$, pool of 198) the infidelity falls below 10^{-2}, 3×10^{-3}, 10^{-3} and 3×10^{-4} after 66, 101, 149 and 194 operators (11 712, 15 528, 22 816 and 30 000 CNOTs including the Givens network; 2.77×10^{-4} after 200); at sixteen sites (pool of 1 022) it falls below 10^{-2} after 187 operators (68 008 CNOTs) and reaches 9.45×10^{-3} after 200. The CNOT count of a prefix is that of its operators as compiled in Sec. 5. The eight-site state at 149 operators, fed to the Trotterized filter of Sec. 5.3 with its forty time sets, gives $1 - \mathcal{F}_M = 9.8 \times 10^{-4}$, 5.1×10^{-4}, 2.8×10^{-4}, 1.7×10^{-4}, 1.2×10^{-4}, 8.6×10^{-5} and 7.3×10^{-5} for $M = 0$–6 at 2.3×10^4, 4.5×10^4, 6.9×10^4, 9.5×10^4, 1.2×10^5, 1.5×10^5 and 1.7×10^5 CNOTs per accepted preparation.

Table S3: Plaquette refinement $2 \times 2 \to 4 \times 4$ on the open square lattice, Path A with $\mu_{\mathrm{buf}} = 2t$. At $(2,2)$ the coarse ground state is a B_1 singlet with zero overlap on the A_2 triplet ground state of the fine lattice; the last row refines the lowest coarse $(A_2, S = 1)$ state, and Δ_{eff} is taken among the thirty lowest fine eigenstates.

$(N_\uparrow, N_\downarrow)$	U/t	dims	E_{low}	E_{high}	(Γ, S) coarse / fine	p_0	Δ	Δ_{eff}	$p(16)$
(1,1)	0	$16 \to 256$	-6.0000	-6.4721	A_1 / A_1	0.805	1.000	2.236	0.99999
(1,1)	4	$16 \to 256$	-5.8136	-6.2638	A_1 / A_1	0.840	0.792	2.096	0.99999
(1,1)	8	$16 \to 256$	-5.7093	-6.1833	A_1 / A_1	0.845	0.711	2.129	0.99998
(2,2)	4	$36 \to 14\,400$	-9.4142	-10.4197	$(B_1, 0)$ / $(A_2, 1)$	0	0.090	—	3×10^{-27}
(2,2)	4	$36 \to 14\,400$	-9.3429	-10.4197	$(A_2, 1)$ / $(A_2, 1)$	0.647	0.090	1.771	0.99997

Table S4: Sector-level UCCSD on the Galerkin coarse chain ($L = 4$, Eq. (4)) and on the fine chain ($L = 8$) at $(2,2)$: reference, variational and exact energies, infidelity with the exact ground state, number of parameters and CNOT count of the preparation (Givens network plus excitation blocks). The last columns give the overlap of the prolonged UCCSD coarse state and its Path-A refinement at $T = 16/t$.

	coarse $L = 4$ (26 parameters, 1720 CNOTs)				fine $L = 8$ (198 parameters, 30864 CNOTs)				UCCSD coarse $\to$ refined	
U/t	E_{HF}	E_{UCCSD}	E_{FCI}	$1 - F$	E_{HF}	E_{UCCSD}	E_{FCI}	$1 - F$	p_0	$p(16)$
0	-6.23607	-6.23607	-6.23607	$< 10^{-15}$	-6.82295	-6.82295	-6.82295	$< 10^{-14}$	0.77733	0.99997
2	-5.23607	-5.43744	-5.43797	1.8×10^{-4}	-5.71184	-6.02962	-6.03209	4.7×10^{-4}	0.81405	0.99977
4	-4.23607	-4.96957	-4.97657	3.3×10^{-3}	-4.60073	-5.61876	-5.63443	3.1×10^{-3}	0.79991	0.99669
8	-2.23607	-4.51765	-4.55859	1.7×10^{-2}	-2.37850	-5.22167	-5.27797	1.2×10^{-2}	0.75667	0.98234

S5 Trotterized refinement at eight sites

Figure S3 replaces the exact slice propagation at $T = 16/t$, $U = 4t$ by first- and second-order product formulas over the groups $[P(H_{\mathrm{low}} - \mu)P^\dagger, K_{\mathrm{even}}, K_{\mathrm{odd}}, D - \mu]$ for Path A and $[(1 + \Lambda \cos^2) K_{\mathrm{even}}, K_{\mathrm{odd}}, D]$ for Path B. The second-order formula reproduces the exact-slice overlap to within 10^{-3} with $n_s = 128$ steps for Path A ($p = 0.99983$), 256 for Path B with $\Lambda = 8t$ (0.99213) and 512 for $\Lambda = 16t$ (0.99342); first-order formulas do not reach it within 10^{-3} even at 512 steps. The counted circuit of Path A does not exponentiate the embedded coarse term as one block: on a device it is the coarse Hubbard step, i.e. the three conditioned coarse groups $P(D_c - \mu)P^\dagger$, $PK^c_{\mathrm{even}}P^\dagger$ and $PK^c_{\mathrm{odd}}P^\dagger$, which do not commute. Repeating the second-order refinement with the coarse term split accordingly (six groups in all) gives $p = 0.997299$, 0.999818, 0.999941 and 0.999943 at $n_s = 64$, 128, 256 and 512, against 0.997456, 0.999831, 0.999941 and 0.999943 with the coarse term as one block; the step count $n_s = 128$ and the cost of Path A are unchanged, and the split-group state is the one used in Table 2. The ramp's product-formula version reproduces its exact-slice overlap with 64 second-order steps at $T = 8/t$ (0.996891 against 0.996861) and gives 0.999659 (exact 0.999885) with 128 steps at $T = 16/t$.

The embedded coarse term of Path A at eight sites is the coarse Hubbard step on the bonding register — 24 rotations, 32 CNOTs unconditioned — with every rotation conditioned on the eight antibonding qubits being empty. With the Gray-code phase-polynomial convention of paper I (exact for few controls but growing as 2^{c+1}) this is 1.2×10^4 CNOTs per application; with a clean-ancilla ladder of relative-phase Toffolis ($6(c - 1) + 2$ extra CNOTs per rotation) it is 1 088, the figure used throughout. In the block Jordan–Wigner ordering the $Z_j Z_{j+L_f}$ terms are the only non-local ones; the site-major ordering, which makes them adjacent at the price of weight-three hopping strings, is the alternative to be compared on a linear-connectivity device.

S6 The exact-ensemble filter

Figure S4 shows the exact ensemble infidelity, Eq. (8), with $\sigma = 5/\Delta_{\mathrm{eff}}$ for the inputs of Sec. 4.5. Applied to the Path-B state at its ceiling ($\Lambda = 16t$, $T = 32/t$, $1 - p \simeq 2 \times 10^{-3}$) the filter reaches 2–4 $\times 10^{-5}$ after six cycles. The exact-ensemble values lie below the sampled, Trotterized ones of Table 2 once the latter approach the Trotter floor, which Eq. (8) does not contain (for the ramp state at $T = 16/t$: 5.3×10^{-6} exact against 6.2×10^{-5} sampled at $M = 6$).

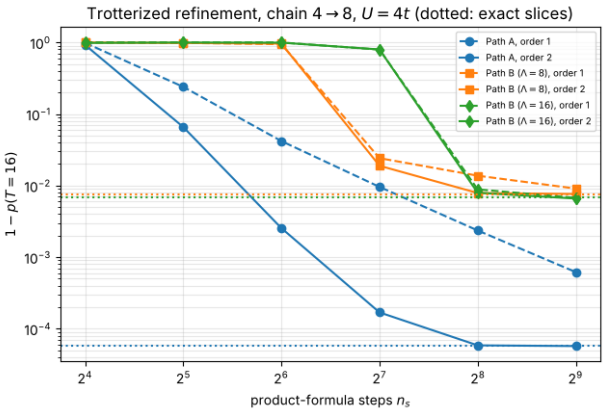


Figure S3: Trotterized refinement of the chain $4 \to 8$ at $U = 4t$, $T = 16/t$: $1 - p$ versus the number of product-formula steps for Path A and Path B ($\Lambda = 8t$, $16t$), first (dashed) and second order (solid); dotted lines are the exact-slice values.

S7 Ingredients of the thirty-two-qubit study

Product formulas through the Kronecker structure. In the $(2, 2)$ sector of the sixteen-site chain (14 400 states) the dense machinery of the eight-site study is unavailable: a $14\,400^2$ complex step unitary cannot be stored, let alone diagonalized. Every term of the product formula is instead applied through Eq. (2). A hopping layer $K_{\text{layer}} = K_\uparrow \otimes \mathbb{1} + \mathbb{1} \otimes K_\downarrow$ acts on ψ reshaped as a 120×120 matrix as $e^{-icK_\uparrow}\, \Psi\, e^{-icK_\downarrow^T}$, with the two 120×120 exponentials obtained from one diagonalization per species; the diagonal interaction is an elementwise phase; and the embedded coarse term of Path A, $P\,X\,P^\dagger$ with X one of the three 784×784 conditioned coarse groups, is applied exactly as $\psi \mapsto \psi + P\big[V(e^{-ic\lambda} - 1)V^\dagger\big]P^\dagger\psi$, with $X = V\lambda V^\dagger$ diagonalized once. The second-order step is the symmetric product of these factors. The ramp uses the groups $[K_{\text{even}}, K_{\text{odd}}, D]$ with the interaction coefficient $\sin^2(\pi s/2)$, Path B the same groups with the even (intra-block) layer scaled by $1 + \Lambda\cos^2(\pi s/2)$, and Path A the six groups $[P(D_c - \mu)P^\dagger, PK^c_{\text{even}}P^\dagger, PK^c_{\text{odd}}P^\dagger, K_{\text{even}}, K_{\text{odd}}, D - \mu]$ with the $\cos^2/\sin^2$ coefficients of Eq. (6). The exact-slice transports use 48 slices with a sparse Krylov exponential. Applied to the eight-site chain, where the dense machinery exists, the Kronecker implementation reproduces the ramp, the split-group Path-A refinement and the filter tracks of Sec. 5.3 to 10^{-13}.

The Trotter-effective ground state. The fixed-step filter projects, in the long run, on the eigenvector of the step unitary $U_{\delta t}$ closest to the ground state, and its eigenphase is the objective energy E_{obj}. At eight sites that eigenvector was read off a Schur decomposition of the step; at sixteen sites it is obtained by iterating the quantized-time filter itself: starting from the exact ground state, $\psi \mapsto \frac{1}{2}(\psi + e^{iE_{\text{obj}}n\delta t}U^n_{\delta t}\psi)$ with n drawn from the Gaussian ensemble of cycle times and E_{obj} updated at every iteration to the eigenphase estimate $-\arg\langle\psi|U_{\delta t}|\psi\rangle/\delta t$, converges in a few tens of iterations to the fixed point of the filter, which is that eigenvector. At eight sites the procedure reproduces the Schur result ($E_{\text{obj}} = -5.620907$, floor 5.8×10^{-5}) to the quoted digits; at sixteen sites, with $\sigma = 36.2/t$ and $\delta t = 0.1/t$, it gives $E_{\text{obj}} - E_0 = +4.92 \times 10^{-3}t$ and a floor of 3.7×10^{-5}, close to the eight-site floor at the same step, as expected for a local error per step that is the same at both sizes.

Gate counts at sixteen sites. The second-order step of the sixteen-site chain costs 304 CNOTs (736 controlled), the Givens network of the free-fermion determinant 2×120 rotations, 480 CNOTs, the prolongation $8 \to 16$ 32 CNOTs, and the coarse term of Path A — the eight-site Hubbard step, 48 rotations, conditioned on the sixteen antibonding qubits with the relative-phase Toffoli ladder — 4 856 CNOTs per application, applied twice per second-order step. The preparation of the coarse eigenstate itself is counted, for every input that starts from it, at the cost of the cheapest eight-site route of Table 2 that reaches 3×10^{-4}, the ramp at $T = 16/t$ without filtering (18 544 CNOTs); its residual infidelity (3.4×10^{-4}) is below the targets of the sixteen-site comparison and is neglected.

Step scans. With the step scans of the notebook (second order, $n_s = 32$ to 512, accepted when the overlap agrees with the exact-slice value to 10^{-3} or has stopped changing): the ramp needs $n_s = 128$ at $T = 16/t$ (0.96927 against 0.96970; 0.91802 at $n_s = 64$) and 256 at $T = 32/t$ (0.99556 against 0.99591; 0.00545, 0.94463 and 0.99304 at $n_s = 32, 64, 128$ — a step of $\delta t = 1/t$ destroys the transport); Path B with $\Lambda = 8t$ needs 256 at $T = 16/t$ (0.99476 against 0.99492; 4×10^{-5}, 0.266 and 0.98688 at $n_s = 32$, 64, 128, the intra-block

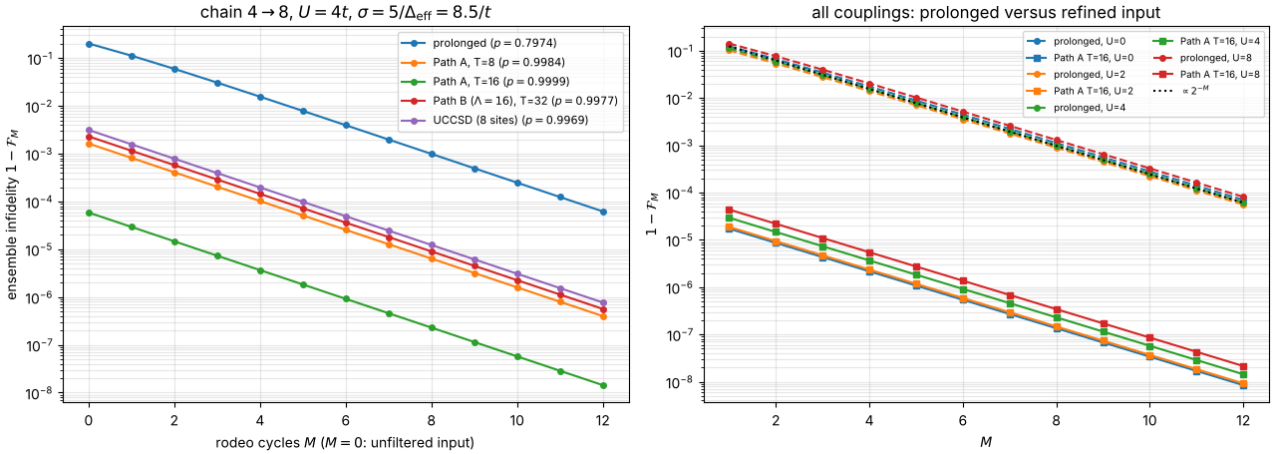


Figure S4: Exact ensemble infidelity of the rodeo filter, Eq. (8), with $\sigma = 5/\Delta_{\text{eff}}$ versus the number of cycles. Left, $U = 4t$: prolonged input, Path-A inputs refined for $T = 8/t$ and $16/t$, the Path-B input at its ceiling and the eight-site UCCSD state. Right: prolonged (dashed) and Path-A $T = 16/t$ (solid) inputs for all four couplings against the 2^{-M} law.

generator of norm $1 + \Lambda$ requiring $\delta t \lesssim 0.06/t$), and is not Trotterized at $T = 32/t$, where it sits at the same ceiling and would need 512 steps; Path A with $\mu_{\text{buf}} = 2t$ and the coarse groups split needs $n_s = 128$ at $T = 16/t$ (0.99835 against 0.99931) while at $T = 8/t$ it plateaus at 0.99079 from $n_s = 128$ on (0.98885, 0.99069, 0.99078, 0.99079, 0.99079 at $n_s = 32$–512) against the exact 0.99237, a residual of 1.6×10^{-3} that no step count removes — the product formula with the coarse term split into non-commuting groups follows a slightly different path, whose end point differs from the exact one by more than the tolerance when the transport is this short.

Sector gap along the ramp. The sector gap of $H_R(s)$ falls monotonically from $0.165t$ at $s = 0$ to $0.051t$ at $s = 1$ ($0.125t$, $0.078t$, $0.057t$ at $s = 0.25, 0.5, 0.75$); the ramp from the determinant is governed, like Path A, by the gap in the singlet, reflection-even sector, which is why both transports converge at all with a sector gap of $0.05t$.

S8 Operating points

Table S5 gives, for the eight-site comparison of Table 2, the minimum CNOTs per accepted preparation with which each input reaches a target ensemble infidelity, with the number of cycles that attains it; the corresponding operating points at sixteen sites are the last four columns of Table 3, read from the twenty time sets with relative standard errors of 10–35% at the cycle counts involved; Table S6 gives the sixteen-site data in full.

References

[1] S. Jansen, M.-B. Ruskai, and R. Seiler, J. Math. Phys. **48**, 102111 (2007).

Table S5: Operating points at eight sites: the minimum CNOTs per accepted preparation with which each input of Table 2 reaches a target ensemble infidelity, with the number of cycles M that attains it in parentheses ($M \leq 6$; a dash means the target is not reached within six cycles).

input	10^{-3}	$3{\times}10^{-4}$	10^{-4}
prolonged / determinant	—	—	—
UCCSD pruned	$6.2{\times}10^{4}$ (2)	$1.2{\times}10^{5}$ (4)	—
UCCSD full	$7.8{\times}10^{4}$ (2)	$1.3{\times}10^{5}$ (4)	—
ramp, $T = 8$	$5.6{\times}10^{4}$ (2)	$1.1{\times}10^{5}$ (4)	$1.5{\times}10^{5}$ (6)
ramp, $T = 16$	$1.9{\times}10^{4}$ (0)	$4.0{\times}10^{4}$ (1)	$9.1{\times}10^{4}$ (3)
Path B, $\Lambda = 8t$	$1.4{\times}10^{5}$ (4)	$1.6{\times}10^{5}$ (5)	—
Path A	$3.0{\times}10^{5}$ (0)	$3.0{\times}10^{5}$ (0)	$3.4{\times}10^{5}$ (2)

Table S6: The sixteen-site matched comparison of Table 3 in full: ensemble infidelity (bootstrap relative standard error in parentheses) and CNOTs per accepted preparation after $M = 0$, 2, 4 and 6 cycles, and the cumulative acceptance after six.

input	p	$M = 0$		$M = 2$		$M = 4$		$M = 6$		
		$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	$1 - \mathcal{F}$	CNOTs	P_6
determinant	0.7162	$2.8{\times}10^{-1}$	480	$8.9{\times}10^{-2}$ (6%)	$5.5{\times}10^{5}$	$2.7{\times}10^{-2}$ (10%)	$1.1{\times}10^{6}$	$7.1{\times}10^{-3}$ (18%)	$1.5{\times}10^{6}$	0.717
prolonged	0.8692	$1.3{\times}10^{-1}$	18 576	$3.9{\times}10^{-2}$ (6%)	$5.2{\times}10^{5}$	$1.1{\times}10^{-2}$ (16%)	$1.0{\times}10^{6}$	$2.4{\times}10^{-3}$ (13%)	$1.4{\times}10^{6}$	0.870
UCCSD pruned (255)	0.9930	$7.0{\times}10^{-3}$	87 776	$1.9{\times}10^{-3}$ (10%)	$5.6{\times}10^{5}$	$5.6{\times}10^{-4}$ (15%)	$1.1{\times}10^{6}$	$1.3{\times}10^{-4}$ (12%)	$1.5{\times}10^{6}$	0.993
ADAPT (187 op.)	0.9901	$9.9{\times}10^{-3}$	68 008	$2.6{\times}10^{-3}$ (8%)	$5.4{\times}10^{5}$	$7.2{\times}10^{-4}$ (14%)	$1.0{\times}10^{6}$	$1.7{\times}10^{-4}$ (11%)	$1.4{\times}10^{6}$	0.990
ramp, $T = 16$, $n_s = 128$	0.9693	$3.1{\times}10^{-2}$	39 392	$7.7{\times}10^{-3}$ (17%)	$5.2{\times}10^{5}$	$2.0{\times}10^{-3}$ (19%)	$1.0{\times}10^{6}$	$4.2{\times}10^{-4}$ (35%)	$1.4{\times}10^{6}$	0.970
ramp, $T = 32$, $n_s = 256$	0.9956	$4.4{\times}10^{-3}$	78 304	$1.1{\times}10^{-3}$ (21%)	$5.5{\times}10^{5}$	$3.2{\times}10^{-4}$ (20%)	$1.0{\times}10^{6}$	$8.6{\times}10^{-5}$ (23%)	$1.4{\times}10^{6}$	0.996
Path B ($\Lambda = 8t$), $T = 16$, $n_s = 256$	0.9948	$5.2{\times}10^{-3}$	96 400	$1.4{\times}10^{-3}$ (7%)	$5.7{\times}10^{5}$	$4.3{\times}10^{-4}$ (13%)	$1.1{\times}10^{6}$	$1.4{\times}10^{-4}$ (13%)	$1.5{\times}10^{6}$	0.995
Path A (split), $T = 16$, $n_s = 128$	0.9983	$1.7{\times}10^{-3}$	1 300 624	$4.3{\times}10^{-4}$ (17%)	$1.8{\times}10^{6}$	$1.7{\times}10^{-4}$ (25%)	$2.3{\times}10^{6}$	$7.4{\times}10^{-5}$ (17%)	$2.7{\times}10^{6}$	0.998